

RE-GPS: Reflective Evolutionary Gradient Prompting System for Large Language Models

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Abstract

Despite their effectiveness in text generation, Large Language Models (LLMs) remain highly sensitive to input instructions (prompts), where suboptimal prompts frequently induce hallucinations, shallow reasoning, and logical inconsistencies. Automatic prompt optimization (autoprompting) based on evolutionary algorithms effectively frames prompt engineering as a black-box optimization problem, yet existing methods still suffer from critical limitations, including susceptibility to local optima and insufficient exploration of the prompt space.

We propose RE-GPS (Reflective Evolutionary Gradient Prompting System), a novel autoprompting framework that integrates textual gradients into a reflective evolution pipeline. By generating precise, data-driven feedback on prompt performance over training subsets, RE-GPS substantially improves short-term reflections that guide the evolutionary search. Evaluations with GPT-4o-mini across 7 diverse datasets demonstrate that RE-GPS achieves state-of-the-art average performance relative to evolutionary baselines while exhibiting strong robustness across tasks.

CCS Concepts

• Computing methodologies → Natural language processing.

Keywords

Reflective Evolution, Prompt Engineering, Autoprompting, Evolutionary Algorithms, Software Engineering

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1 Introduction

Large Language Models (LLMs) demonstrate remarkable capabilities across diverse natural language tasks, particularly in software engineering, where they serve as powerful assistants for code generation, automated test creation, and intelligent code refactoring, significantly boosting developer productivity [1]. Yet, despite their scale, they frequently lack deep problem comprehension and remain notoriously sensitive to input instructions (prompts), often

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producing reasoning failures and hallucinations under suboptimal guidance [3]. To alleviate this, automated prompt generation (autoprompting) based on Evolutionary Algorithms (EAs) replaces laborious manual engineering with algorithmic search [2]. However, their methods have several limitations, rely predominantly on undirected mutation strategies, severely limiting their efficiency and search space exploration. Frameworks like EvoPrompt [4] and GEPA [5] often suffer from premature convergence to local optima due to "blind" trial-and-error mechanisms. Unlike alternative approaches, PromptBreeder [6] lacks stagnation escape when the population converges, while, ReflectivePrompt produces blind reflections without task-specific feedback, remaining vulnerable to initial descriptions.

To address these limitations, we introduce the Reflective Evolutionary Gradient Prompting System (RE-GPS), a novel framework unifying reflective evolution with textual gradients [7]. First, RE-GPS grounds the evolutionary search by constructing a dynamic problem description, synthesizing the vague initial prompt with empirical training examples. Second, it transforms standard crossover into a precise, error-correcting operation via short-term reflections; these are generated by extracting *textual gradients* from an explicit analysis of K specific failure instances for each parent. Third, to prevent premature convergence, RE-GPS continuously aggregates these epoch-level insights into a long-term reflection memory, utilizing this accumulated historical knowledge to rigorously steer elitist mutation.

2 RE-GPS Framework

Framework Overview. The RE-GPS approach formulates prompt engineering as a guided black-box optimization, its pipeline is illustrated in Figure 1. Since user instructions are often vague, RE-GPS first constructs a comprehensive problem description by synthesizing the initial prompt with empirical examples sampled from the training dataset. This ensures the LLM captures task-specific dependencies and output formats. Using this enriched description as context in the meta-prompt, the system initializes the starting population by generating N_{pop} diverse rephrasings of the original prompt.

Evolutionary Loop. In each epoch, RE-GPS systematically refines the prompt candidates through a multi-stage process. It begins with *Parent Selection*, where the system selects N_{pop} parent pairs with probabilities linearly proportional to their empirical fitness scores. Next, in *Textual Gradients*, for each parent prompt, RE-GPS isolates K training examples where the prompt failed; the LLM acts as a critic to generate a *textual gradient*—detailed natural-language reasoning about the prompt's flaws with optimizations tailored to the data distribution [7]. Subsequently, during *Short-term Reflection*, for each parent pair, the system generates a short-term reflection by synthesizing the parent prompts with their respective textual gradients, capturing localized, data-driven optimization strategies. Then,

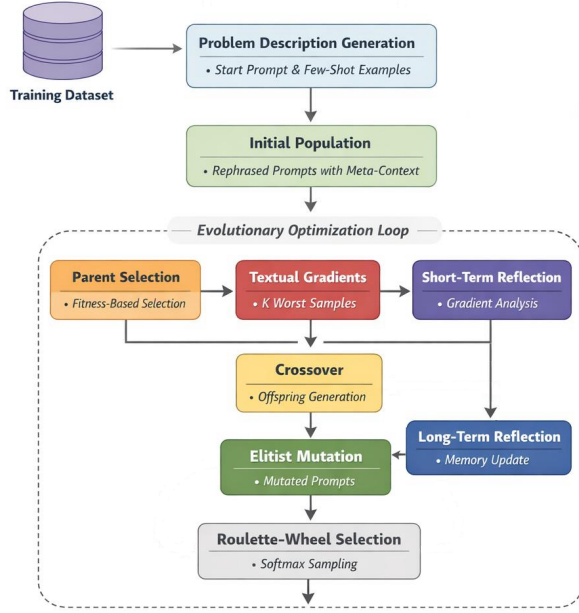


Figure 1: The overall pipeline of RE-GPS. The process initializes with an example-driven problem description. Each evolutionary epoch incorporates textual gradients to form short-term reflections for crossover, while continually updating a long-term reflection memory to guide elitist mutation.

in *Gradient-Guided Crossover*, guided by these short-term reflections, a crossover operation merges parent traits to yield N_{pop} new semantically refined offspring prompts. After that, *Long-term Reflection* accumulates population-level knowledge: RE-GPS updates its overarching long-term reflection text by synthesizing its previous state with the current epoch’s short-term reflections. Finally, *Elitist Mutation*, based strictly on the current elite (best-performing) prompt and the renewed long-term reflection, applies elitist mutation to generate an additional N_{pop} diversified prompts.

Final Population Selection. To consolidate the newly generated candidates into a fixed-size population, RE-GPS calculates their fitness and applies a softmax function to transform these objective scores into survival probabilities. A *Roulette-Wheel Selection* method then samples the final N_{pop} prompts. This probabilistic approach efficiently balances the exploitation of high-performing instructions with the preservation of structural diversity.

3 Experiments and Results

To evaluate the capabilities of RE-GPS, we conducted experiments across a diverse set of language tasks.

Experimental Setup. We conducted a comparative evaluation of RE-GPS against existing state-of-the-art evolutionary prompt optimizers on seven well-established datasets, covering a diverse range of NLP tasks and evaluation metrics: **CommonGen**: Natural Language Inference (BERTScore); **XSUM**: Text Summarization (BERTScore); **MediQA**: Medical Question Answering (BERTScore);

GSM8K: Mathematical Reasoning (Exact Match); **SQuAD v2**: Reading Comprehension / QA (BERTScore); **TweetEval**: Sentiment Classification (Macro F1-Score); and **CONCODE**: Code Generation (CodeBERTScore).

All evaluations and prompt generation procedures were conducted using the gpt-4o-mini model to ensure a consistent computational baseline.

Results. Our experimental evaluation demonstrates that RE-GPS effectively generalizes across varied tasks. As illustrated in Table 1, RE-GPS achieves the highest average metric value across all competitive baselines.

Table 1: Performance comparison of automated prompt generation methods using gpt-4o-mini. Best results are bolded.

Dataset	EvoPrompt	PromptBreeder	Ref.Prompt	GEPA	RE-GPS (Ours)
CommonGen	0.808	0.803	0.808	0.823	0.824
XSUM	0.714	0.711	0.721	0.716	0.725
MediQA	0.736	0.734	0.720	0.720	0.720
GSM8K	0.905	0.892	0.861	0.880	0.912
SQuAD v2	0.827	0.859	0.860	0.922	0.912
TweetEval	0.557	0.576	0.575	0.475	0.561
CONCODE	0.664	0.655	0.655	0.670	0.671
Average	0.744	0.747	0.744	0.743	0.761

The integration of textual gradients enables our approach to generalize effectively across diverse NLP tasks, along with strong robustness to context degradation.

4 Conclusion

In this paper, we proposed RE-GPS, a novel automated prompt optimization framework based on reflective evolution and textual gradients. By grounding self-reflections in actual error analysis driven by training data, RE-GPS drastically improves the quality of evolutionary operations. Evaluated across 7 distinct datasets encapsulating various NLP disciplines, RE-GPS demonstrated superior robustness and achieved the highest average performance metrics among the tested baselines.

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